SmartDaaS — TRIPOD Reporting Checklist

Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis Or Diagnosis (TRIPOD)

Manuscript: Real-World Validation of Machine Learning Models for HIV Treatment Adherence Prediction and Care Gap Quantification: A Multi-Country Analysis of 192,732 Clinical Records

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Repository: github.com/Kchinthala15/smartdaas-hiv-validation · Zenodo DOI: 10.5281/zenodo.20676666

TRIPOD reference: Moons KGM et al. Ann Intern Med. 2015;162(1):W1-W73.

Item	Checklist item	Reported	Section / location in manuscript
TITLE AND ABSTRACT			
1	Title and abstract — Identify the study as developing and/or validating a multivariable prediction model, the target population, and the outcome to be predicted.	Yes	Title + Abstract (Section 5). Title explicitly names 'Validation,' 'HIV Treatment Adherence Prediction,' and 'Multi-Country Analysis.'
INTRODUCTION			
2	Background and objectives — Explain the medical context and rationale for developing or validating the model, including references to existing models.	Yes	Sections 1–2. Marcus et al. (2020), Krakower et al. (2019), Zhao et al. (2025) cited as existing models.
METHODS			
3	Source of data — Describe the study design or source of data for development and validation datasets separately.	Yes	Section 3.1.1 (Quality of Care HIV dataset — development);

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			Sections 3.1.2 and 3.8 (CEPHIA and temporal holdout — validation).
4a	Participants — Specify key elements of the study setting (e.g. primary care, general population), separately for development and validation.	Yes	Section 3.1.1: 27,288 HIV-positive patients on ART across multiple facility types in the Nigerian national HIV programme.
4b	Participants — Describe eligibility criteria for participants.	Yes	Section 3.1.1: HIV-positive patients enrolled on ART with sufficient follow-up to determine outcome status.
4c	Participants — Give details of treatments received, if relevant.	Yes	Section 3.1.1: all patients receiving ART; regimen type included as contextual variable.
5a	Outcome — Clearly define the outcome that is predicted by the model, including how and when assessed.	Yes	Sections 3.1.1 and 3.4: outcome defined as ArvAdherenceLatestLevel = 'Poor' at most recent clinical assessment. Model card (GitHub) documents clinical pathway.
5b	Outcome — Report any actions taken to blind assessment of the outcome to be predicted.	Yes	Section 3.3 (Data Leakage Prevention): outcome determined from records subsequent to predictor measurement; no

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			prospective contamination.
6a	Predictors — Clearly define all predictors, including how and when measured.	Yes	Section 3.2 (Feature Engineering): all 15 features defined with derivation logic. Full implementation in utils.py (GitHub).
6b	Predictors — Report any actions taken to blind assessment of predictors.	Yes	Section 3.3: train/test split performed before any model fitting; all preprocessing fitted on training folds only.
7	Sample size — Explain how the study size was arrived at.	Yes	Section 3.1.1: full dataset of 27,288 patients used. Events Per Variable = 62.7 (940 events / 15 predictors) — well above the minimum of 10.
8	Missing data — Describe how missing data were handled, with details of any imputation method.	Yes	Sections 3.4 and 4.1: missingness documented by feature and facility type (CD4: 17–21%). SMOTE applied to training folds only; test/validation sets untouched.
9a	Statistical analysis — Describe how predictors were handled in the analyses.	Yes	Section 3.2: continuous predictors retained in continuous form; Random Forest handles non-linearity and interactions

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			without distributional assumptions.
9b	Statistical analysis — Specify type of model, all model-building procedures, and method for internal validation.	Yes	Section 3.4: Random Forest classifier selected; 10-fold stratified CV; SMOTE within training folds only. Section 3.7: ablation analysis.
9c	Statistical analysis — For validation, describe how predictions were calculated.	Yes	Section 3.8 (Temporal Validation): temporal split by ART initiation date (pre/post September 2016). External validation on PHIA and CEPHIA datasets described.
9d	Statistical analysis — Specify all measures used to assess model performance.	Yes	Section 3.4: AUC-ROC, sensitivity, specificity, Brier score, calibration curves, DCA. Table 1 presents cross-model comparison (DeLong z-test, p<0.001).
9e	Statistical analysis — Describe any model updating arising from the validation.	Yes	Section 3.5: Platt scaling and isotonic regression calibration. Local recalibration design documented in model card (GitHub).
10	Risk groups — Provide details on how risk groups were created, if done.	Yes	Section 3.10 (DCA): threshold range 0.01–0.50. Section 3.11: subgroup strata

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			defined by sex, age, CD4, WHO stage.
11	Development vs. validation — For validation, identify any differences from development data in setting, eligibility, outcome, and predictors.	Yes	Section 3.8: temporal split described explicitly. Differences in time period noted. PHIA external validation population characteristics described.
RESULTS			
12a	Participants — Describe the flow of participants through the study, including numbers with and without the outcome.	Yes	Section 4.1: 27,288 total; 940 poor-adherence events (3.4%). Training/holdout/temporal splits all described with sample sizes.
12b	Participants — Describe characteristics of participants including number with missing data for predictors and outcome.	Yes	Section 4.1: mean age 35.2 years; 61.4% female; CD4 distribution; WHO stage at ART start; missingness by facility type.
12c	Participants — For validation, show comparison with development data distribution.	Yes	Section 3.8: temporal holdout distribution compared to training set. External PHIA population described.
13a	Model development — Specify number of participants and outcome events in each analysis.	Yes	Table 1: n=27,288 (CV); n=10,540 (holdout); n=19,084

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			training / n=8,179 test (temporal validation).
13 b	Model development — Report results from any model selection process.	Yes	Table 1: four classifiers compared. Random Forest selected on AUC-ROC (DeLong z-test, p<0.001 vs. Logistic Regression).
14 a	Model specification — Present the full prediction model to allow individualised predictions.	Yes	Full code at github.com/Kchinthala15/smartdaas-hiv-validation. Feature importances in Section 4.5 and Figure 5.
14 b	Model specification — Explain how to use the prediction model.	Yes	GitHub README.md and MODEL_CARD.md provide full deployment guidance. Live platform: smartdaas-hiv-validation.onrender.com.
15	Model performance — Report performance measures with confidence intervals.	Yes	Sections 4.2–4.3: AUC 0.9753 ±0.0015 (CV); AUC 0.9727 (95% CI: 0.970–0.975, holdout); sensitivity 87.3% (95% CI: 86.4–88.2%); specificity 95.7% (95% CI: 95.2–96.2%); Brier score 0.079.
16	Model performance — Report results from any model updating arising from the validation.	Yes	Sections 3.5 and 4.3: calibration curves and Brier score reported for holdout and temporal

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			validation. Platt scaling and isotonic regression results in Figure 2, Panel B.
DISCUSSION			
17	Limitations — Discuss any limitations of the study.	Yes	Section 5.4: temporal validation performance gap (0.975 CV vs 0.772 temporal), single-source training data, SMOTE synthetic examples, point-estimate economic projections all discussed.
18a	Interpretation — For validation, discuss results with reference to development data and any other validation data.	Yes	Section 5.1: CV vs holdout vs temporal AUC compared. Zhao et al. (2025, AUC ~0.94) cited as published benchmark.
18b	Interpretation — Give an overall interpretation of the results.	Yes	Sections 5.1–5.3: interpretation contextualised against published ML-HIV literature and clinical implications discussed.
19	Implications — Discuss potential clinical use of the model and implications for future research.	Yes	Sections 5.2 (Clinical Implications) and 5.5 (Future Directions): operational use case described; priority future work listed.
OTHER INFORMATION			

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20	Supplementary information — Provide information about availability of supplementary resources (protocol, datasets, code).	Yes	Section 8 (Data availability): full code and data at github.com/Kchinthala15/smartdaas-hiv-validation. Live platform: smartdaas-hiv-validation.onrender.com.
21	Funding — Give the source of funding and role of funders.	Yes	Declarations: No external funding received. Independent researcher.

Notes

All 21 TRIPOD items are reported as present in the manuscript or supplementary GitHub repository. No items are absent or not applicable. This checklist is available as a supplementary file alongside the manuscript.

TRIPOD items most commonly missing in published ML prediction models — missing data handling (Item 8), calibration (Item 9e), confidence intervals on performance (Item 15), and validation updating (Item 16) — are all fully addressed in this manuscript. The most recent systematic review of ML models for HIV treatment interruption prediction (Kwarah et al., 2025, BMC Global and Public Health) found 75% of 12 models had high risk of bias and only 2 of 12 included any external validation. Several models published in 2025–2026 have since added external validation, but all remain single-country studies.

To our knowledge, SmartDaaS is among the first identified HIV treatment-retention prediction platforms combining multi-country external validation with calibration analysis, decision curve analysis, subgroup fairness assessment, per-patient SHAP explainability, and a live deployed platform compatible with standard EMR exports, requiring no new infrastructure, with local recalibration support — a methodological and operational combination not previously identified in the published literature. Note: external validation was conducted using cross-sectional population-based PHIA survey data as a proxy validation across multiple country contexts; full external validation on longitudinal programme EMR data with identical outcome definitions is planned as the primary objective of the APIN and AMPATH pilot studies.